



UNIVERSITY OF CREATIVE TECHNOLOGY CHITTAGONG

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Project Report

On

**AI Driven Smart Rain & Water-Level Monitoring and Alert System
using ESP32 & Thingspeak**

Artificial Intelligence

CSE-321

Submitted by:

Mohammad Ashraful Islam

ID: 240321022

Md. Ali

ID: 230321004

CSE(R1), 6th Semester

Submitted to:

Professor Dr. Shahadat Hossain Sir

Lecturer

Department of Computer Science & Engineering

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PROJECT REPORT

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1. Introduction

Flooding and unseasonal rainfall pose a recurring threat to homes, farmland, and low-lying infrastructure. Manual monitoring of water levels and rainfall is slow, inconsistent, and often reaches the observer too late to act. This project presents a low-cost, microcontroller-based monitoring system built around the ESP32 development board that continuously tracks rainfall, ambient temperature and humidity, and the distance to a nearby water surface (used as a proxy for water level). When pre-defined thresholds are crossed, the system raises an immediate local alert through a buzzer and an indicator LED, giving the user an early warning before a situation becomes critical.

2. Objectives

1. To detect rainfall in real time using a rain/moisture sensor.
2. To measure the distance to a water surface using an ultrasonic sensor, as an indicator of rising water level.
3. To continuously monitor ambient temperature and humidity for environmental context.
4. To trigger an audible and visual alert when unsafe conditions are detected.
5. To implement the entire system on a single, low-power ESP32 microcontroller.

3. System Overview

The system is built around the ESP32, which acts as the central processing and decision-making unit. It reads data from three sensors — an HC-SR04 ultrasonic distance sensor, an FC-37 rain sensor, and a DHT11 temperature/humidity sensor — and uses this data to decide when to activate the buzzer and LED alert outputs. The ESP32's built-in Wi-Fi also allows the system to be extended in the future to push live data and alerts to a mobile app, web dashboard, or messaging service.

3.1 Components Used

- ESP32 Development Board — central controller with Wi-Fi/Bluetooth
- HC-SR04 Ultrasonic Sensor — measures distance to water surface (water-level proxy)
- FC-37 Rain Sensor — detects presence and intensity of rainfall
- DHT11 Sensor — measures ambient temperature and humidity

- Buzzer — audible alert output
- Red LED (with 330 Ω resistor) — visual alert output
- Voltage divider (resistors) — steps down the HC-SR04 Echo signal to safe 3.3 V logic
- Jumper wires, breadboard, and 5 V power supply

4. Circuit / Pin Connection Diagram

The table below lists every physical connection between each sensor/actuator and the ESP32 GPIO pins, exactly as wired in this implementation.

4.1 HC-SR04 Ultrasonic Sensor (Water Level)

Component	Component Pin	ESP32 Pin
HC-SR04	VCC	VIN (5V)
HC-SR04	GND	GND
HC-SR04	TRIG	GPIO 5
HC-SR04	ECHO	GPIO 18 (via voltage divider)

Note: The Echo pin outputs a 5 V signal, but ESP32 GPIOs are rated for 3.3 V. A resistor voltage divider on the Echo line is mandatory to avoid damaging the GPIO pin.

4.2 FC-37 Rain Sensor

Component	Component Pin	ESP32 Pin
FC-37	VCC	VIN (5V)
FC-37	GND	GND
FC-37	DO (Digital Out)	GPIO 14
FC-37	AO (Analog Out)	GPIO 34 (optional)

Note: DO gives a simple rain/no-rain digital flag; AO (on an ADC-capable pin) optionally provides a graded reading of rainfall intensity.

4.3 DHT11 Temperature & Humidity Sensor

Component	Component Pin	ESP32 Pin
DHT11	VCC	3V3
DHT11	GND	GND
DHT11	DATA	GPIO 4

4.4 Alert Outputs — Buzzer and LED

Component	Component Pin	ESP32 Pin
Buzzer	Positive (+)	GPIO 13
Buzzer	Negative (–)	GND
Red LED	Anode (long leg)	GPIO 2 via 330 Ω resistor
Red LED	Cathode (short leg)	GND

5. Working Principle

- The ESP32 triggers the HC-SR04 sensor periodically and measures the echo return time to calculate distance to the water surface.
- The FC-37 sensor's digital output is polled to detect whether rain is currently falling; its analog output can additionally be read for intensity.
- The DHT11 sensor is read at regular intervals to log ambient temperature and humidity.
- The ESP32 compares the measured distance and rain status against configured threshold values (e.g., water level rising above a safe distance, or rain detected continuously for a set duration).
- If a threshold is breached, the ESP32 drives GPIO 13 high to sound the buzzer and GPIO 2 high to light the LED, alerting anyone nearby.
- Once conditions return to normal, the alerts are automatically switched off.

6. Applications

- Early flood warning for homes, basements, and low-lying farmland
- Automated rainwater harvesting tank-level monitoring
- Smart agriculture — irrigation scheduling based on rainfall and humidity
- Weather-station data logging for small-scale research projects
- Drainage and stormwater monitoring in urban infrastructure

7. Future Scope

- Send real-time alerts to a smartphone via Wi-Fi (e.g., using ESP32's built-in connectivity with a push-notification or messaging service)
- Log sensor data to the cloud for historical trend analysis and graphing
- Add a solar panel and battery for fully off-grid, outdoor deployment
- Integrate a water pump relay to automatically drain water once a critical level is reached

8. Conclusion

This project demonstrates a practical, low-cost approach to environmental hazard monitoring using readily available sensors and a single ESP32 microcontroller. By combining ultrasonic distance sensing, rain detection, and temperature/humidity monitoring with simple buzzer and

LED alerts, the system provides timely, automated warning of potentially hazardous rain or water-level conditions — forming a solid foundation that can be extended with wireless connectivity and cloud logging in future iterations.